

Profile M.S. candidate in Chemical Engineering at NTUST, specializing in electrochemistry, ultra-low-temperature electrolytes and lithium battery reliability / aging; co-author on a Nature Energy submission. Research & Industry Long-Life Li Battery Evaluation for Smart Water Meters Partner: a smart-water-meter battery maker Project lead Screening methodology Designed full-stage test protocols, experimental workflows and statistical validation models. Database & SOP Established the test database and standardized high-throughput screening procedures (SOP). Advanced analysis EIS, OCV drip, self-discharge / lifetime estimation, accelerated aging and X-ray CT internal structure analysis. Training & tech transfer Trained QA and R&D teams to independently run large-batch screening and data interpretation. Nature Energy Journal submission — battery thermal stability study Status Under review Co-author Responsible for TGA thermal measurement Performed TGA sample preparation, measurement setup and data interpretation for battery material samples, providing thermal stability data supporting the manuscript's conclusions. Experience 2025 – Present Sustainable Electrochemical Energy Development (SEED) Center, NTUST Lab Equipment Procurement & Asset Maintenance Manager Procurement (>NTD 2M): Led procurement of glovebox O₂ sensors, a DSC press set with annual maintenance contract, a hydraulic electrode punch, vacuum pumps, 10 Eppendorf pipettes, low-temperature baths and TGA platinum pans; also replace instrument consumables. Lab safety: monitor glovebox O₂ / moisture levels and handle sensor faults; manage and inspect gas cylinders, run lab safety walk-throughs and brief newcomers on safety; helped with on-site response to incidents such as a cut from broken glassware and a glovebox catalyst-canister leak. Hazardous materials: handle lithium metal and assemble coin cells in a glovebox, prepare ether / carbonate organic-solvent electrolytes, and dismantle and handle faulty Li-SOCl₂ cells. - Assembled cells and prepared electrodes in a battery dry room, working in cleanroom garments under clean-area gowning and entry procedures. TGA testing service: Ran TGA measurements and data interpretation for graduate students from 6 labs across Chemical, Mechanical and Materials Engineering at NTUST — over 600 samples in total, including my own lab's. Instrument maintenance: Responsible for maintenance and calibration of the key thermal analysis instruments, TGA Q500 and DSC 2500. Summer 2022 New Taipei City Dept. of Urban Development Summer Intern, Housing Development Section - Reviewed rental subsidy eligibility, cross-checking property, income, and insurance records across agencies (tax bureau, land registry, NHI); handled frontline public inquiries and disputes. 2021 – 2024 Tunghai University Student Housing / GEPT Resident Assistant (2 yrs) / Exam Staff (part-time) - Oversaw safety and order in a 100-resident dormitory, mediating disputes and tracking facility safety issues with the dorm supervisor; as part-time exam staff, handled test-paper check-in with zero errors. Conferences & Training Presenter 2026 TwICHe Annual Meeting 2026/12/05–06 • Feng Chia University Title: <i>Weakly Solvating Allyl Ether Electrolytes for Subzero Li SPAN Batteries: Solvation Structure, Transport Properties, and Separator Wettability</i> Vendor seminar TA Instruments ARES-G3 launch & applications forum 2026/07/17 • National Taiwan University Vendor lecture TA Instruments TGA lecture 2026/06/03 • National Taiwan University	Skills & Instruments Electrochemistry & batteries Impedance spectroscopy (EIS) Battery aging analysis Reliability analysis Low-temp electrolytes Characterization Thermal analysis (TGA / DSC) Synchrotron (TLS 20A / TPS 27A / TPS 45A) Rheology (ARES-G3) Viscometer & densimeter (DMA 5002) Process & operations Lithography (nanoimprint / self-assembly) Equipment procurement & maintenance Software Aspen Plus V14 Aspen Dynamics Microsoft Office OriginPro TA Universal Analysis TA TRIOS Education 2025 – 2027 (expected) National Taiwan University of Science and Technology M.S. in Chemical Engineering Focus on low-temperature electrolytes and battery aging. Coursework: Nanomaterials and Applications (lithography, nanoimprint & self-assembly), Process Safety & Radiation Protection. 2020 – 2024 Tunghai University B.S. in Chemical & Materials Engineering Focus on unit operations & process control (Aspen Plus V14 / Aspen Dynamics simulation). EHS Licenses ✓ Dedicated Wastewater and Sewage Treatment Specialist (Class B) Certified 2026 ✓ Indoor Air Quality Maintenance Management Specialist Certified 2026 ✓ Waste Disposal Technician (Class A) Certified 2026 ○ Occupational Safety and Health Management Personnel (Level B) 2026 / 11 • mid-Nov exam scheduled ○ Occupational Safety Management Specialist (Level A) 2027 / 03 • exam scheduled Instruments Maintenance TGA Q500 DSC 2500 Operation ARES-G3 Anton Paar DMA 5002 Glovebox Procurement Glovebox O₂ sensor DSC sample press Hydraulic electrode punch Vacuum pump Eppendorf Pipette Low-temperature bath
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